THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The organelle that packages and dispatches materials in vesicles is the: 1

(a) Golgi apparatus **(b)** ribosome **(c)** mitochondrion **(d)** nucleus

2. Genes are segments of: 1

(a) DNA **(b)** protein **(c)** lipids **(d)** cell wall

3. Which of the following is a unicellular organism? 1

(a) Paramecium **(b)** frog **(c)** mango tree **(d)** earthworm

4. The 'suicide bags' of the cell are the: 1

(a) vacuoles **(b)** ribosomes **(c)** plastids **(d)** lysosomes

5. When a plant cell is kept in a hypotonic solution, it: 1

(a) shrinks **(b)** swells **(c)** does not change **(d)** bursts always

6. The cell membrane is described as selectively permeable because it: 1

(a) is made of cellulose **(b)** allows no substances

(c) allows only certain substances to pass **(d)** allows all substances

7. Protein synthesis takes place on the: 1

(a) lysosomes **(b)** ribosomes **(c)** Golgi bodies **(d)** vacuoles

8. When a plant cell is kept in a hypertonic solution, it: 1

(a) swells **(b)** shrinks (plasmolysis)

(c) bursts **(d)** remains same

9. The movement of particles from a region of higher concentration to lower concentration is: 1

(a) diffusion **(b)** active transport **(c)** osmosis **(d)** filtration

10. The plastids that contain chlorophyll are: 1

(a) lysosomes **(b)** leucoplasts **(c)** chromoplasts **(d)** chloroplasts

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Compare prokaryotic and eukaryotic cells in two points. 2

12. What is the function of lysosomes? 2

13. Differentiate between diffusion and osmosis. 2

14. Why are mitochondria called the powerhouse of the cell? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the structure and function of the cell membrane. 3

16. Differentiate between mitosis and meiosis (three points). 3

17. Name the parts of the nucleus and state the function of each. 3

OR

Differentiate between plant and animal cells in three points. 3

18. Why is the cell called the structural and functional unit of life? 3

OR

Name the organelle responsible for each: 3
(a) respiration
(b) photosynthesis
(c) protein synthesis
(d) digestion of waste
(e) control of the cell.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain diffusion and osmosis with an activity each and the importance in the cell, plants and animals. 5

OR

Explain how the shape and structure of the cells is related to their function with five examples. 5

20. Compare prokaryotic and eukaryotic cells in five points with examples. 5

OR

Describe the nucleus and chromosomes, DNA and genes and their importance in heredity. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND